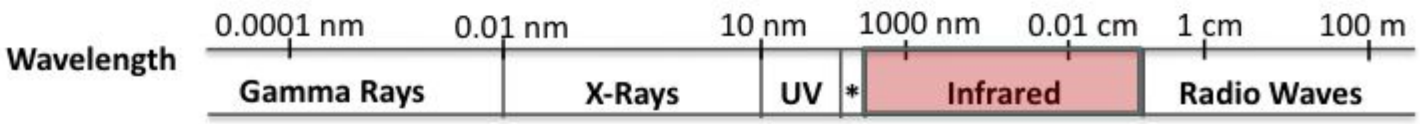
Free-Space Optics (FSO): Advantages and Security Challenges

For high-speed connections (Gbps, up to Tbps), FSO is a promising candidate

- **Using infrared bands →** support extremely high data rates (Gbps, up to Tbps)
- **Low power consumption and License-free spectrum**



Security Challenges in FSO

- **FSO operates over an open nature**
  → Optical signals can be intercepted by eavesdropper

Key question: How can we address security challenges in FSO?

1.Introduction

Traditional encryption (relies on computational hardness)
→risk that it could be broken in the future with the advancement of quantum computers

QKD (ultimate security based on the laws of physics)

→requires expensive equipment

Traditional PLS (Uses channel feature as a source of randomness: like turbulence)

→atmospheric turbulence only changes few mili seconds(T_c)

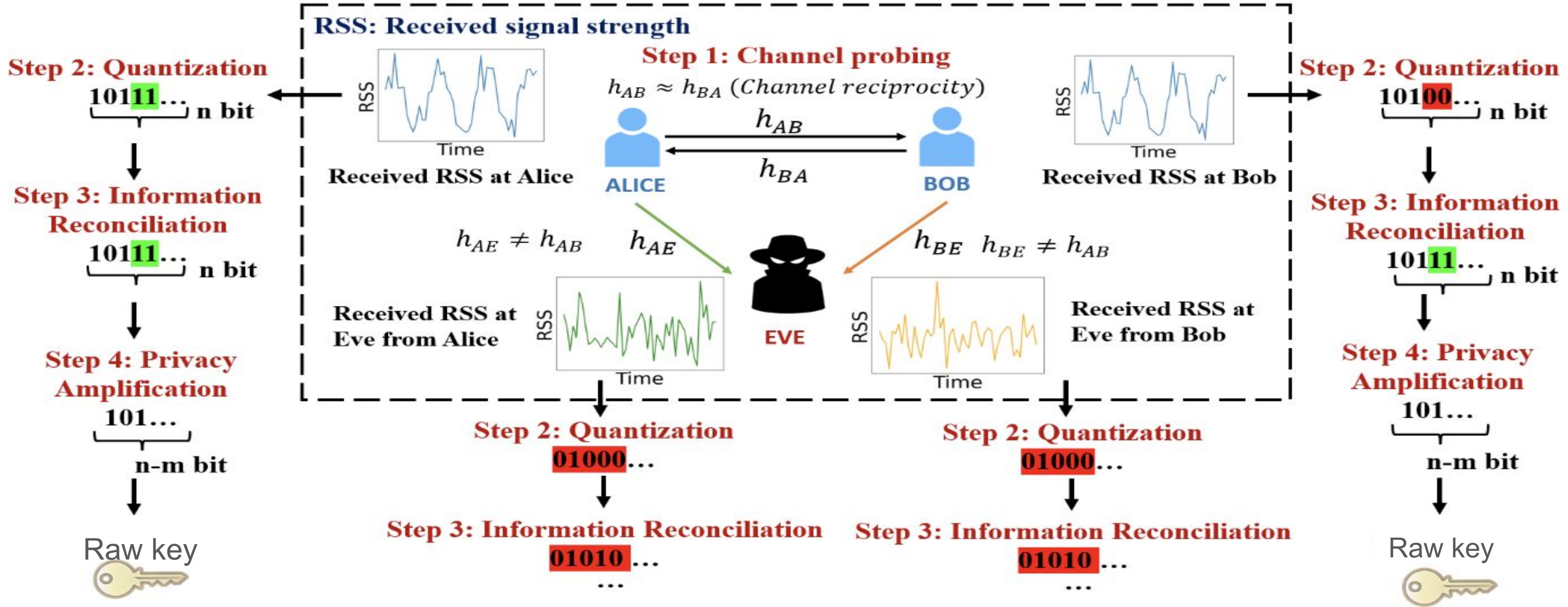
→the key generation rate is far too slow for modern high-speed optical communications

Atmospheric turbulence-controlled cryptosystem

・Using existing CW lasers and receivers, cheaper than QKD

・This system can provide higher secret key rate than traditional PLS

- PLS-KG generates a shared secret key from reciprocal channel randomness without direct key exchange.
  - Step 1: Channel Probing: Alice and Bob exchange pilot signals to observe the random channel properties. If the channel reciprocity exists, the channels h_{AB} and h_{BA} are highly correlated.
  - Step 2: Quantization: The measured values are converted into binary bits (0 and 1).
  - Step 3: Key Reconciliation: Errors are corrected (by ECC) so both keys are identical.
  - Step 4: Privacy Amplification: The key is refined to remove any information an eavesdropper might have.



2.The key idea to create secret key

Alice

Raw Key

00111101…

CSP RNG

Gamma-Gamma Shaping (Inverse CDF)

CDF-Based Quantization (0.5 threshold)

0101010101….

Bob

Raw Key

00111101…

CSP RNG

Gamma-Gamma Shaping (Inverse CDF)

CDF-Based Quantization (0.5 threshold)

0101010101….

Extract seed to initialized CSPRNG
Generate <u>random number sequence</u>

Every Tc seconds (Coherence time of atmosphere), update **α and β (parameter of Gamma-Gamma distribution)**

Secret Key

used for One-Time Pad encryption and then immediately destroyed

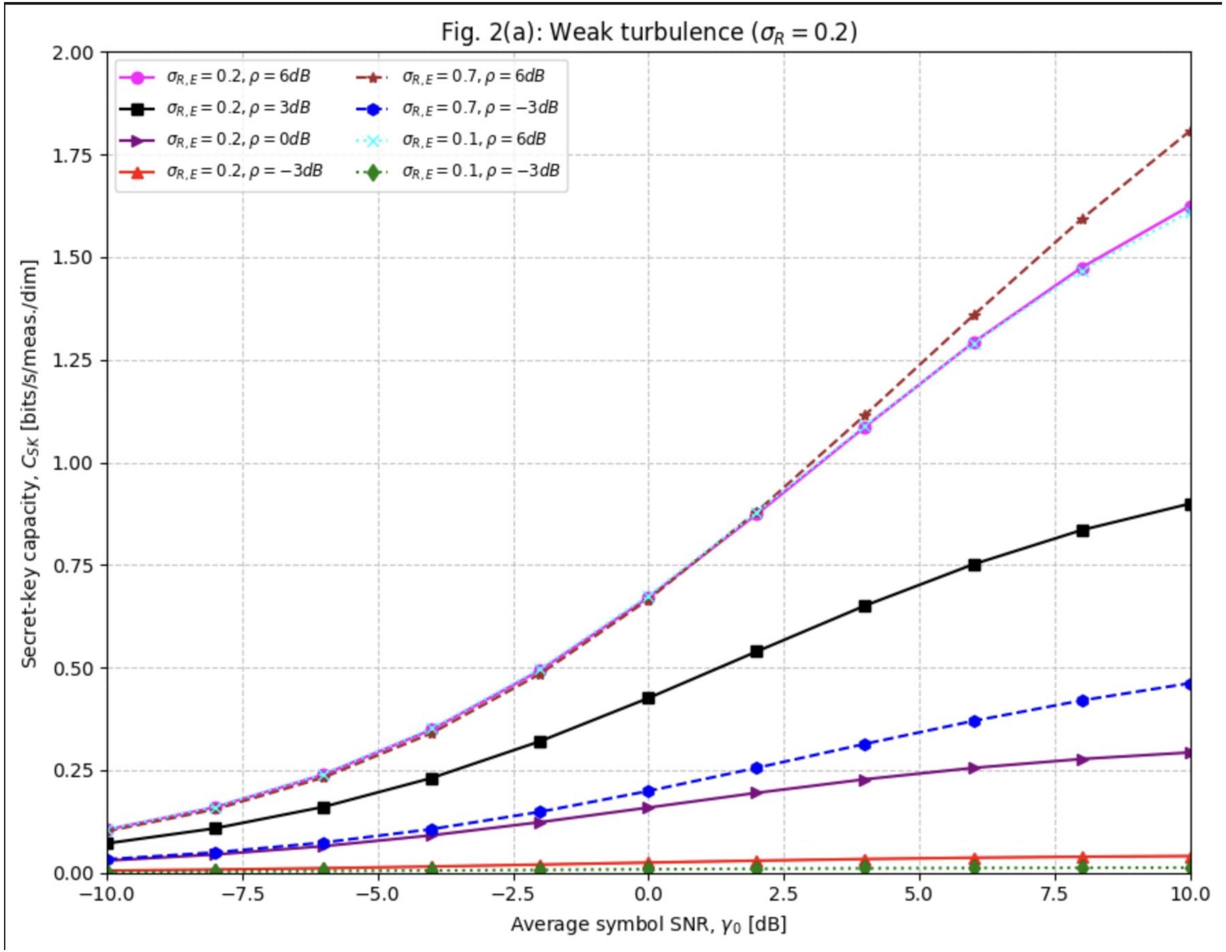
3.The secret-key(common secure sequence) capacity results of initialization phase

σR: turbulence strength between Alice and Bob

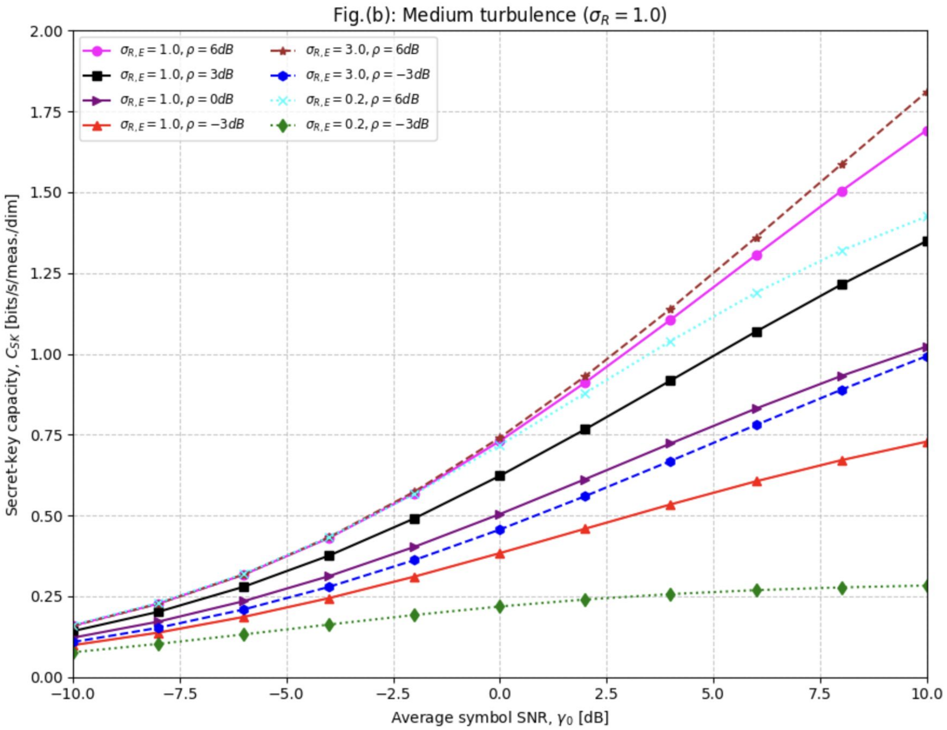
σRE: turbulence strength between Alice and Eve

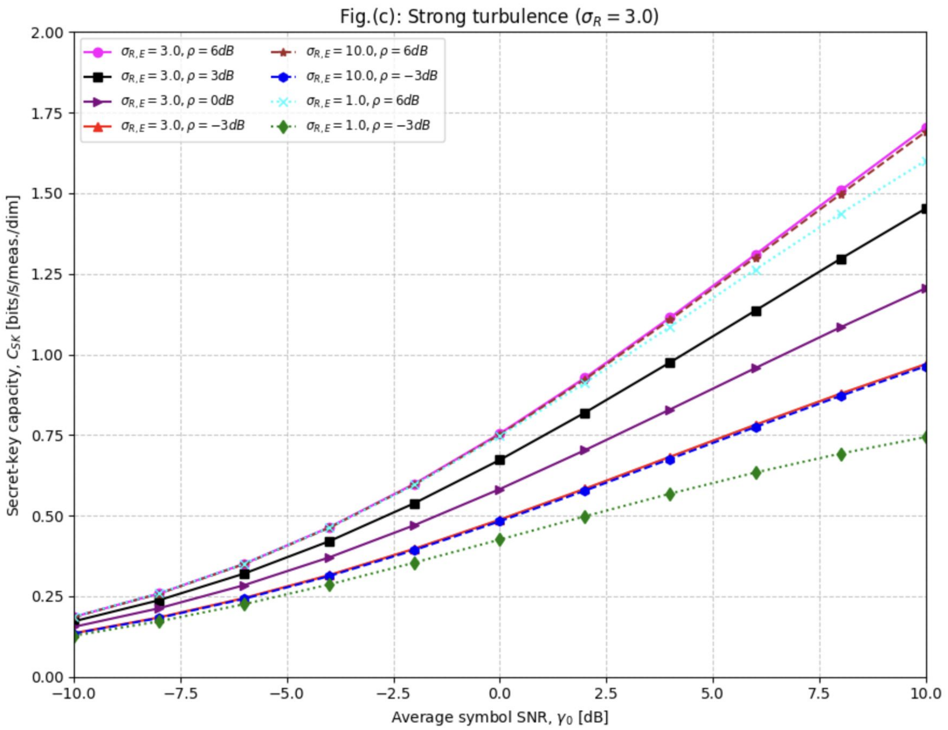
ρ: Bob and Eve's average SNR ratio

If ρ is low(such as -3),
It means Eve has a much better receiver or is located closer to the Alice



3.The secret-key(common secure sequence) capacity results of initialization phase





(b) Medium turbulence σR=1.0

(c) Strong turbulence σR=3.0

4.The secret-key capacity over FSO channel against the SNR in the main channel, when the proposed encryption scheme is used, for different turbulence strengths.